

Assessment of Autoencoder and Transformer Networks for Airfoil/Cylinder Flow Wake and Jet Flow Reconstruction

1 Introduction

The purpose of this report is to compare the reduced-order representation obtained from β -VAE with the classical POD approach. The same framework is applied to flow over cylinder and airfoil, and subsonic and supersonic jet. The analysis follows the reduced-order modeling idea used by Solera-Rico et al. [1], where the β -VAE was shown to provide compact nonlinear latent representations for fluid-flow reconstruction and prediction. The fluctuating velocity field is represented using the streamwise and crosswise velocity components,

$$\mathbf{q}(t) = [u(x, y, t), v(x, y, t)]. \quad (1)$$

The β -VAE maps each flow snapshot into a low-dimensional latent space via encoder and reconstructs the velocity field through the decoder. POD is used as the linear reference method. The reconstruction quality is measured using cumulative reconstructed energy,

$$E = \left[1 - \frac{\left\langle \sum_{i=1}^{N_p} \sum_{j=1}^2 (q_{ij} - \tilde{q}_{ij})^2 \right\rangle}{\left\langle \sum_{i=1}^{N_p} \sum_{j=1}^2 q_{ij}^2 \right\rangle} \right] \times 100\%, \quad (2)$$

where q_{ij} is the reference velocity fluctuation, $\tilde{q}_{ij}$ is the reconstructed value, and $\langle \cdot \rangle$ denotes averaging over snapshots.

For prediction, the trained β -VAE encoder first maps the flow field into latent space. The easy-attention model then predicts the future latent variables, and the decoder reconstructs the predicted velocity field. Therefore, the prediction is evaluated both in latent space and in the reconstructed physical field.

2 Circular-cylinder Flow

The circular-cylinder case is a canonical vortex-shedding problem. The wake is more periodic than the separated airfoil case, so the dominant behavior is expected to be low-dimensional. The β -VAE energy and mode results show that two latent variables are sufficient to represent the primary shedding dynamics. The first two latent variables contain a clear dominant frequency at $f/f_c \approx 0.045$, and their temporal signals are smooth and periodic.

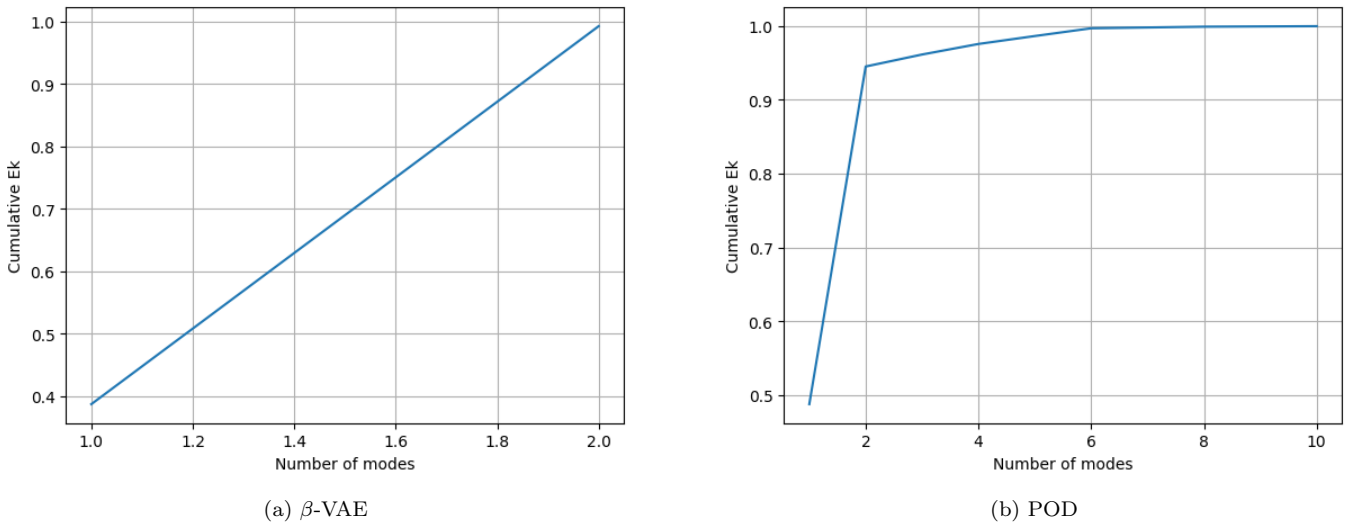


Figure 1: Cumulative reconstructed energy for the cylinder.

The POD result also shows the same behavior. The first POD mode captures approximately 49% of the energy, and the first two modes capture about 95%. Higher modes represent harmonic content and smaller corrections. The

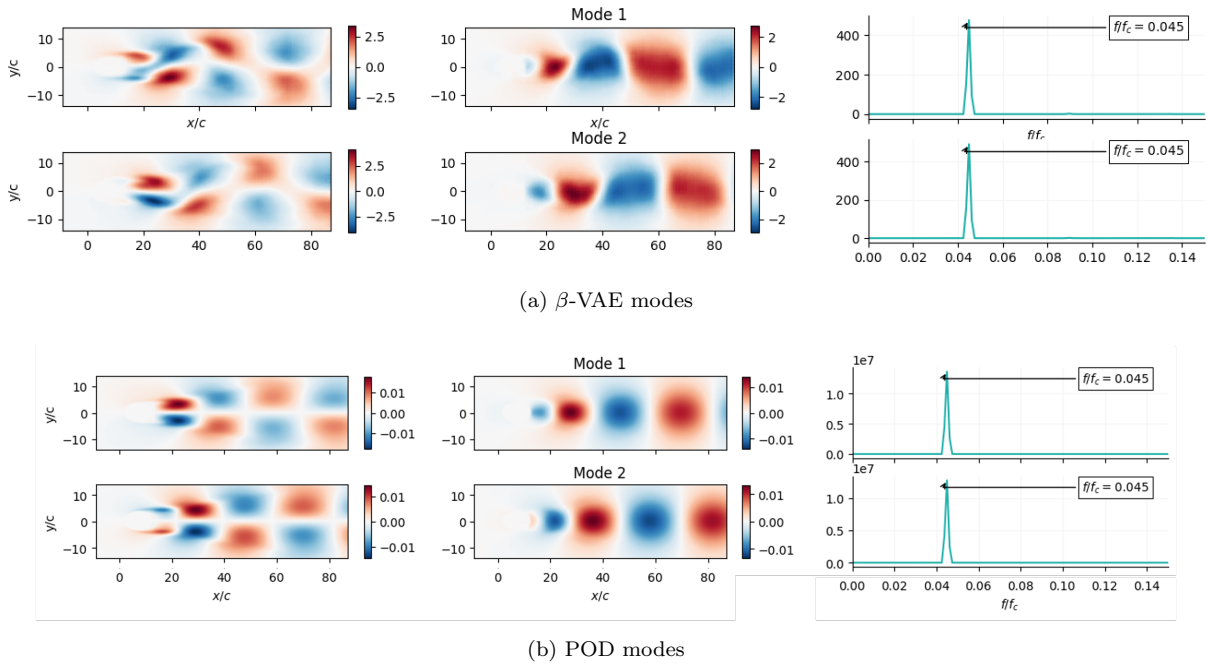


Figure 2: Spatial modes and spectra for the circular-cylinder flow.

first two POD modes have the same dominant frequency as the β -VAE modes, while modes 3–4 and 5–6 correspond to higher harmonics near $f/f_c \approx 0.090$ and 0.135 .

The cylinder prediction results are shown in Fig. 3. At both $t + 65t_c$ and $t + 100t_c$ ($t_c = dt$ obtained from snapshot sampling time), the predicted fields agree closely with the true and reconstructed fields. The streamwise velocity captures the main low-speed wake region behind the cylinder, while the crosswise velocity captures the alternating vortex-shedding structure. The prediction remains accurate at long prediction time because the cylinder wake is governed by a smooth periodic latent-space trajectory.

3 Airfoil at angle of attack (α) = 10°

The airfoil case at $\alpha = 10^\circ$ represents a relatively organized wake compared with the higher-angle case. The β -VAE cumulative reconstructed energy is shown in Fig. 4. The energy increases from about 40% with one latent variable to almost 99% with two latent variables. This indicates that the dominant unsteady behavior of this flow can be represented using only two latent variables.

For POD, the first mode captures about 48% of the energy, while the first two modes capture approximately 95%. The energy approaches nearly 99% after about four modes. Therefore, POD also performs well for this case, but it requires more modes than the β -VAE to reach the same reconstruction level. This is expected because the flow is periodic and can be represented efficiently by mode pairs.

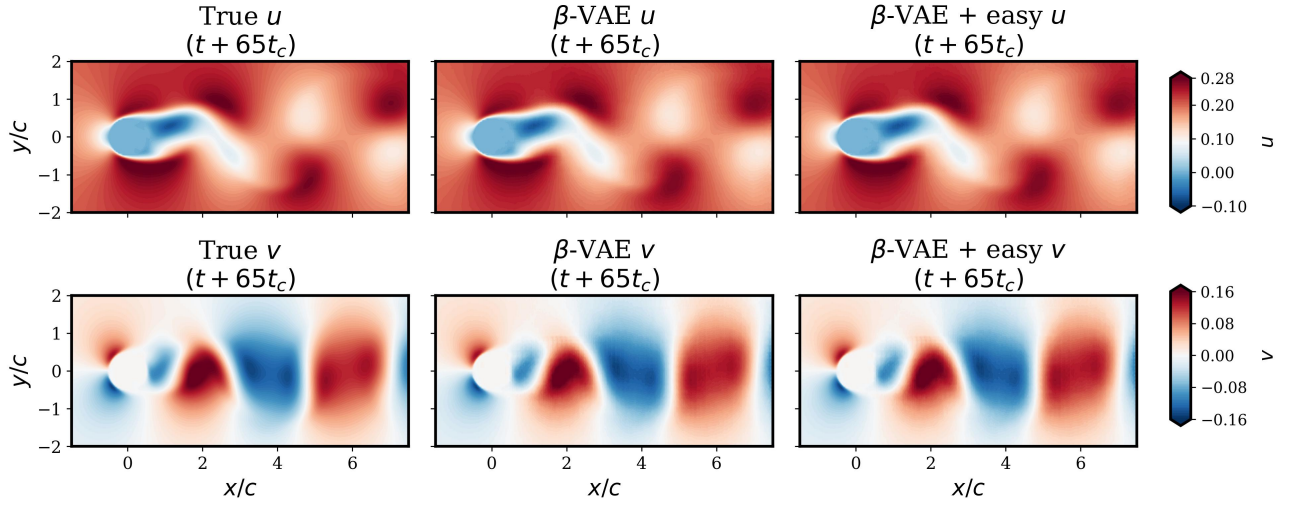
The β -VAE modes are shown in Fig. 5. The two modes show alternating positive and negative wake structures downstream of the airfoil. The spectra show a clear dominant frequency at $f/f_c \approx 0.038$ for both latent variables. This means that the two latent variables represent the same wake-shedding process with a phase shift.

The POD modes also identify the same dominant shedding frequency. The first two POD modes have a peak at $f/f_c \approx 0.038$, while higher POD mode pairs contain harmonic frequencies near $f/f_c \approx 0.078$ and 0.116 . This confirms that both POD and β -VAE capture the same dominant physics. However, the β -VAE represents the primary wake dynamics more compactly.

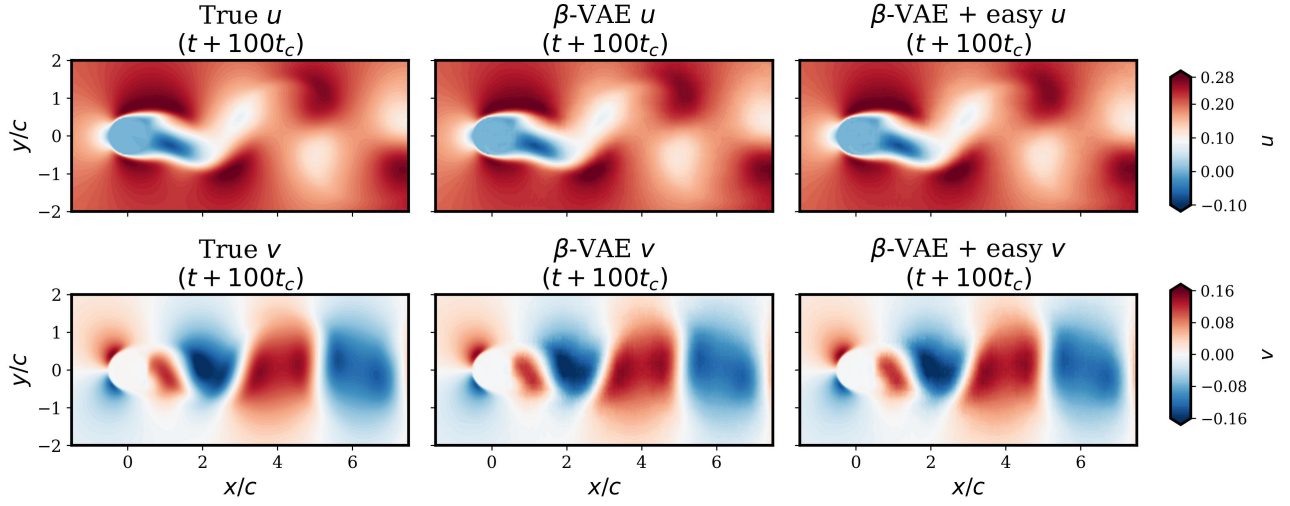
The prediction result for the $\alpha = 10^\circ$ case is shown in Fig. 6. The predicted field from the β -VAE plus easy-attention model closely follows both the true field and the direct β -VAE reconstruction. The streamwise velocity preserves the low-speed wake region and the downstream alternating structures. The crosswise velocity also captures the sign and location of the main vortical structures.

4 Airfoil at angle of attack (α) = 35°

The airfoil case at angle of attack (α) = 35° is more challenging because the flow contains stronger separation and more nonlinear wake dynamics. The β -VAE energy curve in Fig. 7 shows that approximately five latent variables reconstruct

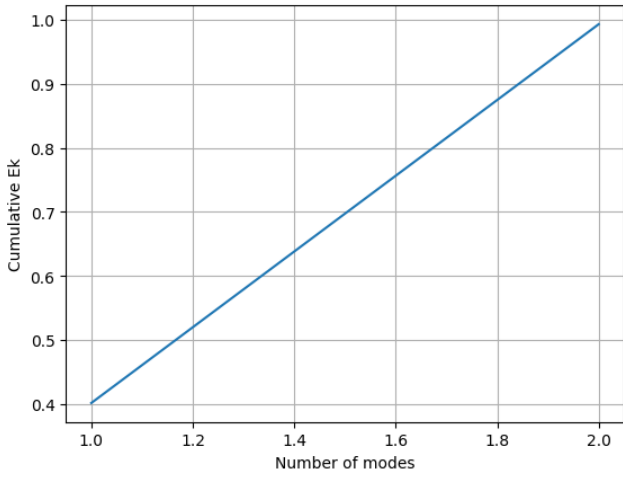


(a) $t + 65t_c$

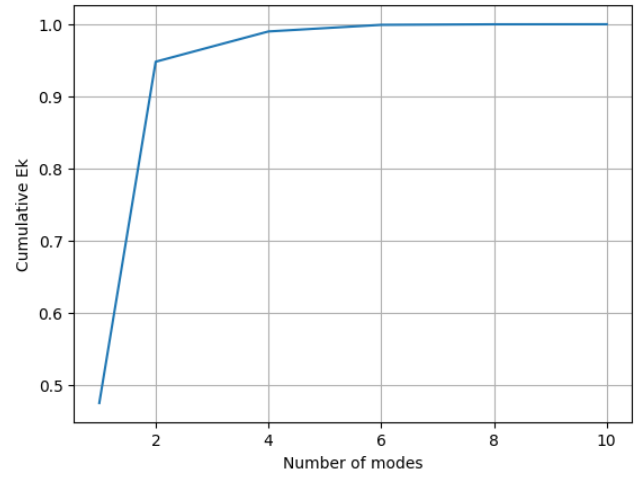


(b) $t + 100t_c$

Figure 3: True, reconstructed, and predicted fields for the circular-cylinder flow.



(a) β -VAE



(b) POD

Figure 4: Cumulative reconstructed energy for the airfoil case at $\alpha = 10^\circ$.

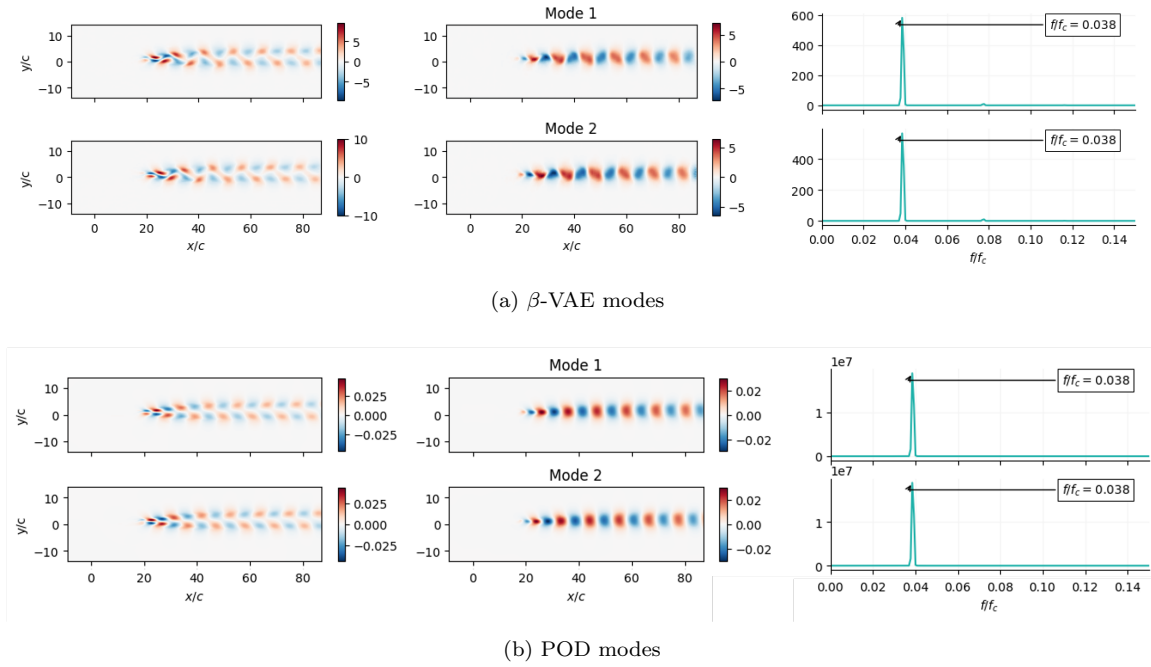


Figure 5: Spatial modes and spectra for the airfoil case at $\alpha = 10^\circ$.

about 74% of the flow-field energy, while about eight latent variables reconstruct nearly the full energy. After this point, additional modes contribute very little. This shows that the β -VAE still provides a compact representation even for a separated and nonlinear wake.

The POD energy curve increases more slowly. With ten POD modes, the reconstructed energy is only about 82%. This shows that POD requires more modes to represent the separated airfoil wake. The difference occurs because POD is a linear decomposition, so nonlinear wake deformation and vortex interaction are distributed over many modes. In contrast, the β -VAE decoder reconstructs the flow through a nonlinear mapping.

The β -VAE and POD modes are compared in Fig. 8. The β -VAE modes contain large-scale coherent structures in the separated wake. The dominant frequency content is concentrated at low nondimensional frequencies, mainly around $f/f_c \approx 0.003$ – 0.004 . These modes represent large-scale, low-frequency separated-flow motion.

The POD modes also capture coherent wake structures, but the flow information is spread over more modes. Some modes represent large-scale wake motion, while higher modes act as corrections to reconstruct the separated flow. This is consistent with the slower growth of the POD energy curve.

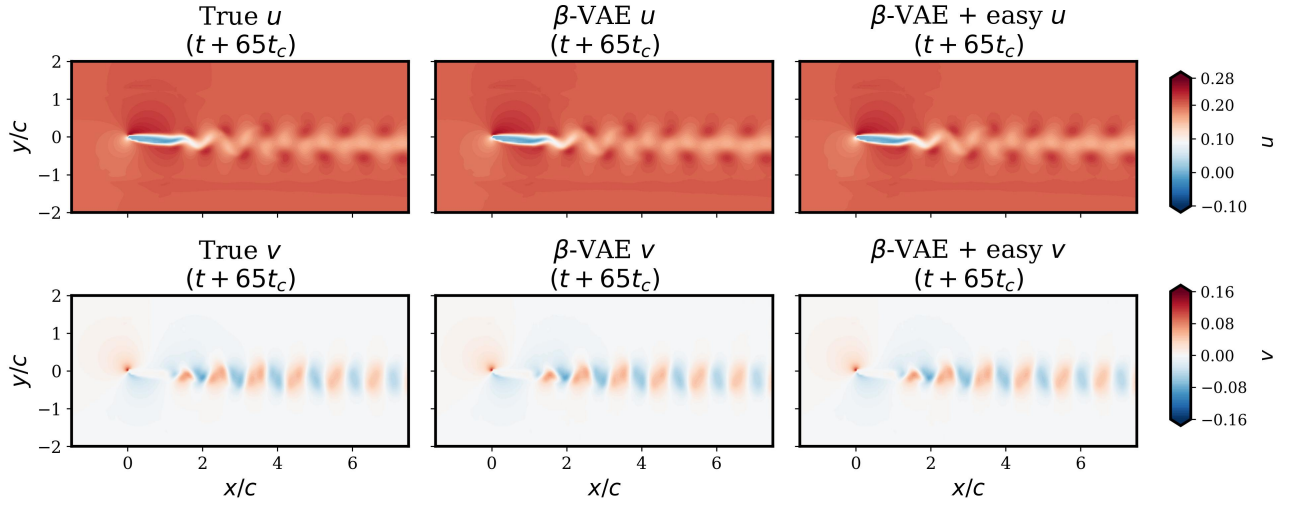
The prediction result for the $\alpha = 35^\circ$ case is shown in Fig. 9. The direct β -VAE reconstruction agrees well with the true field, showing that the decoder can reconstruct the separated wake from the latent variables. The predicted field also preserves the main wake structures.

5 Subsonic jet at Mach number (M) = 0.6

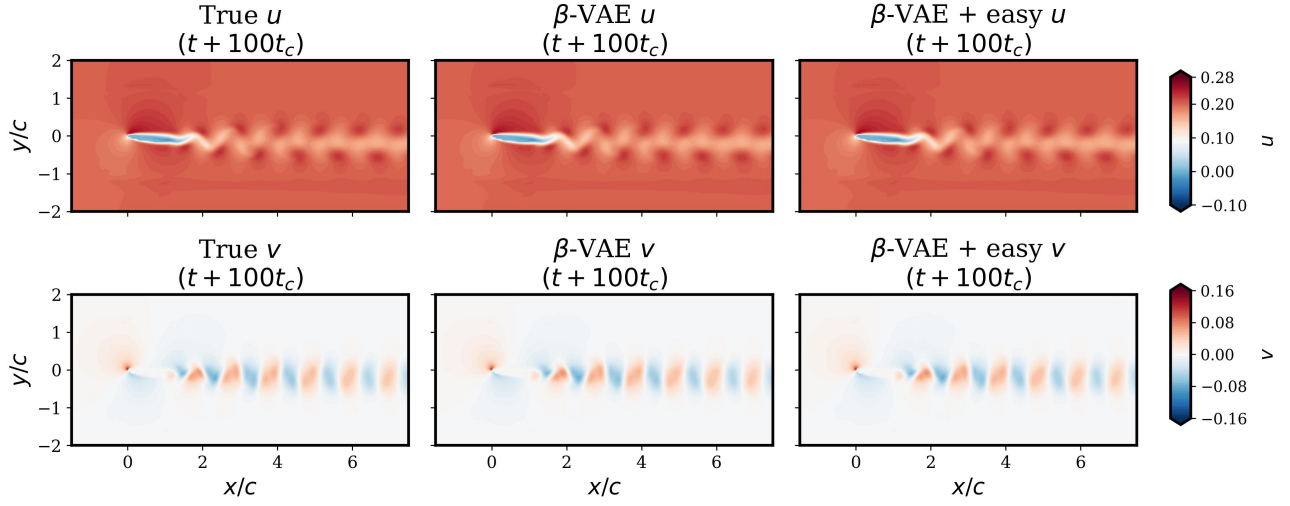
The subsonic jet case at $M = 0.6$ is used to assess the ability of the β -VAE to represent a shear-layer-dominated free jet. Unlike the cylinder and low-angle airfoil cases, the jet contains spatially developing Kelvin–Helmholtz instability, vortex roll-up, downstream jet spreading, and small-scale fluctuations in the mixing layer. Therefore, this case is a more challenging test for the reduced-order model because the dominant structures are not fixed in space and evolve as the jet develops downstream.

The cumulative reconstructed energy for the subsonic jet is shown in Fig. 10. The β -VAE reconstruction energy increases rapidly with the number of latent variables and then gradually approaches a plateau. This shows that, based on the reconstructed-energy metric, most of the recoverable flow-field information is captured using a limited number of latent variables using β -VAE. In comparison, the POD reconstruction energy increases more gradually over the same modal range. Therefore, for the present subsonic jet dataset, the β -VAE achieves a higher reconstructed energy than POD for the same number of retained modes.

The spatial modes shown in Fig. 11 further illustrate the difference between the β -VAE and POD representations. The β -VAE modes show localized positive and negative structures within the jet region, with the strongest variations appearing in the downstream portion of the domain. The POD modes also show alternating positive and negative

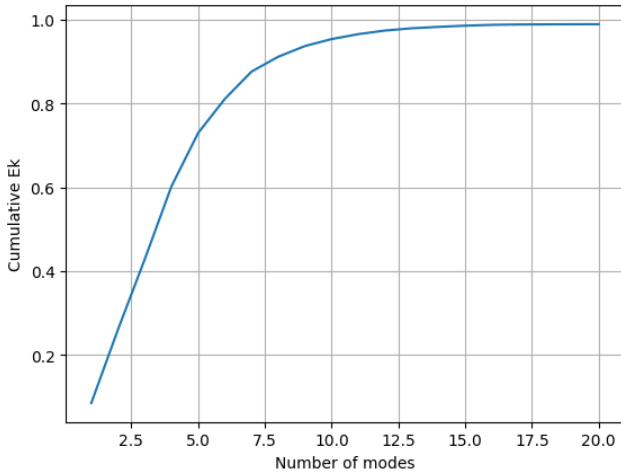


(a) $t + 65t_c$

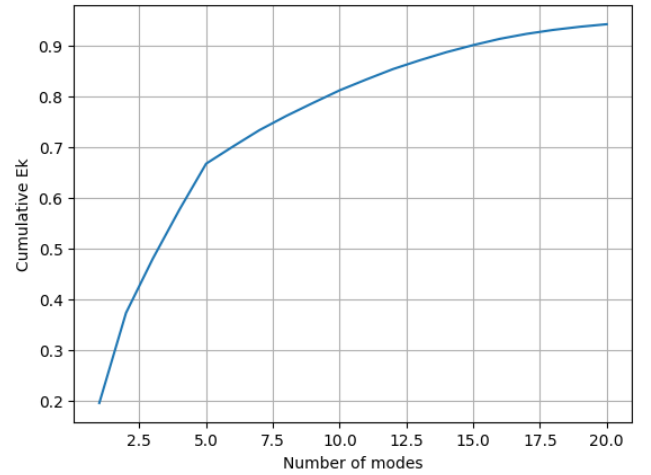


(b) $t + 100t_c$

Figure 6: True, reconstructed, and predicted fields for the airfoil case at $\alpha = 10^\circ$.

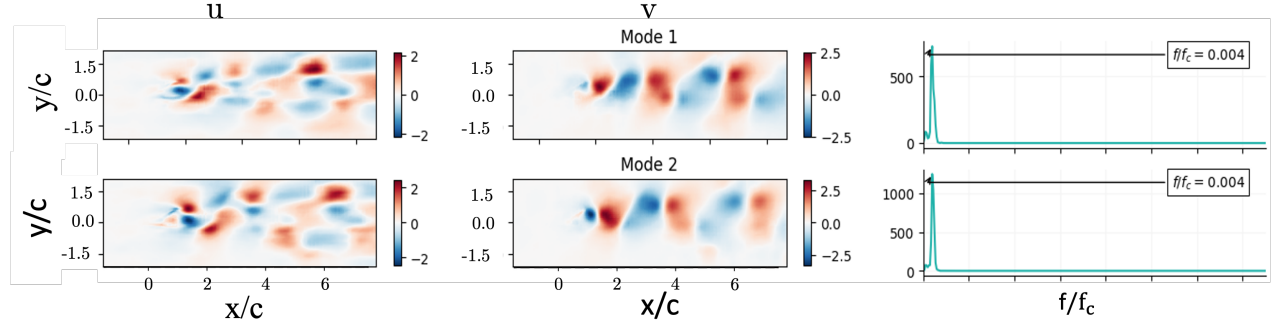


(a) β -VAE

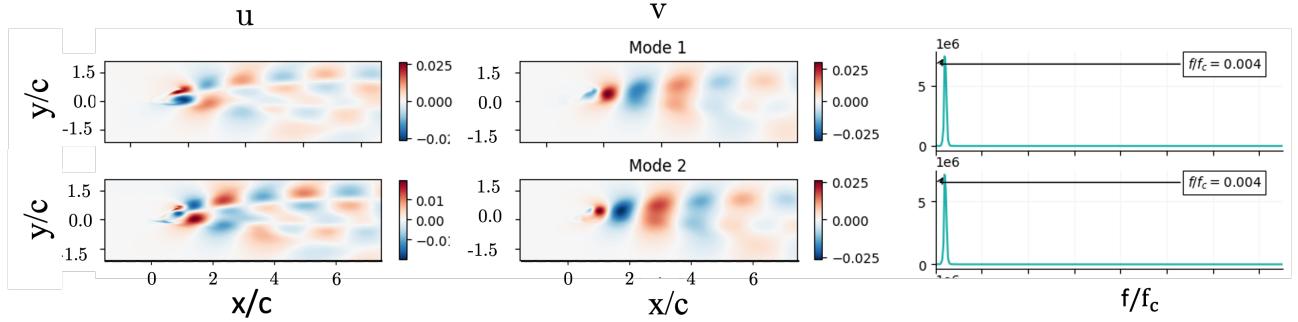


(b) POD

Figure 7: Cumulative reconstructed energy for the airfoil case at $\alpha = 35^\circ$.



(a) β -VAE modes



(b) POD modes

Figure 8: Spatial modes and spectra for the airfoil case at $\alpha = 35^\circ$.

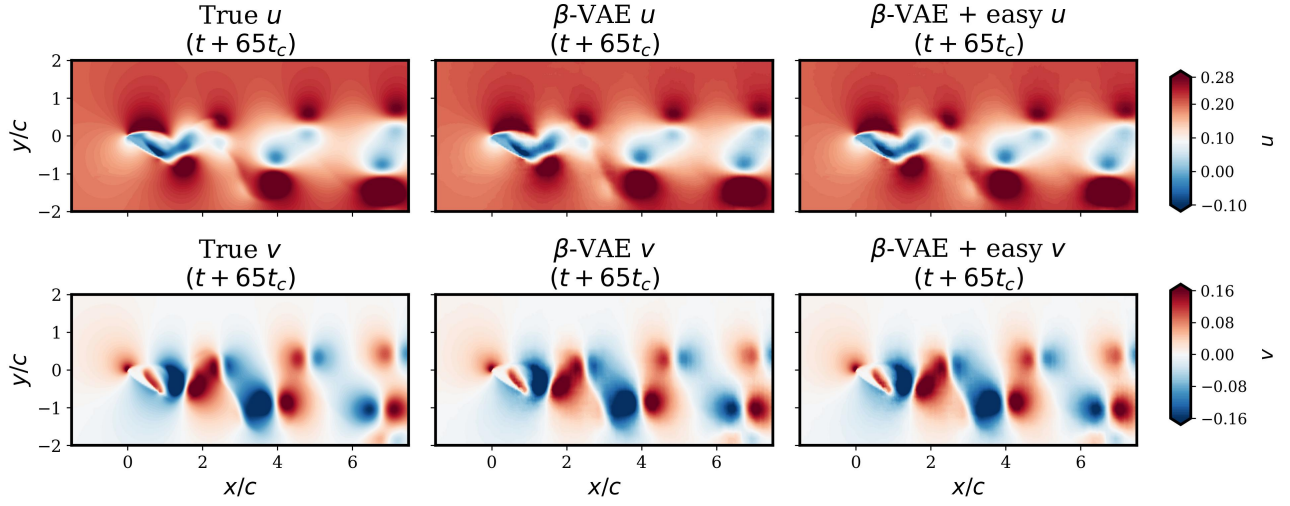
regions, but the structures appear distributed differently across the retained modes. Based on these mode visualizations, the β -VAE and POD both capture spatial variations in the jet field, but their modal representations are not identical. Together with the reconstructed-energy result in Fig. 10, the comparison suggests that the β -VAE provides a more compact representation for the present subsonic jet dataset.

The prediction results in Fig. 12 compare the true vorticity magnitude field, the direct β -VAE reconstruction, and the β -VAE plus easy-attention prediction at $t+65t_c$ and $t+199t_c$. At both prediction times, the β -VAE reconstruction preserves the main structure of the subsonic jet. The shear layers are reconstructed, and the large-scale regions of high vorticity are located in approximately the same downstream region as the true solution. The reconstructed field also captures the progressive growth of the jet width and the dominant vortex patterns in the mixing layer. The β -VAE plus easy-attention prediction follows the direct β -VAE reconstruction closely. This indicates that the temporal prediction model can evolve the latent variables in a physically consistent manner for the subsonic jet. The predicted field preserves the main vorticity pattern and the downstream distribution of coherent structures. However, some differences are visible in the detailed shape and intensity of the vortical regions, especially farther downstream, where the flow becomes more complex.

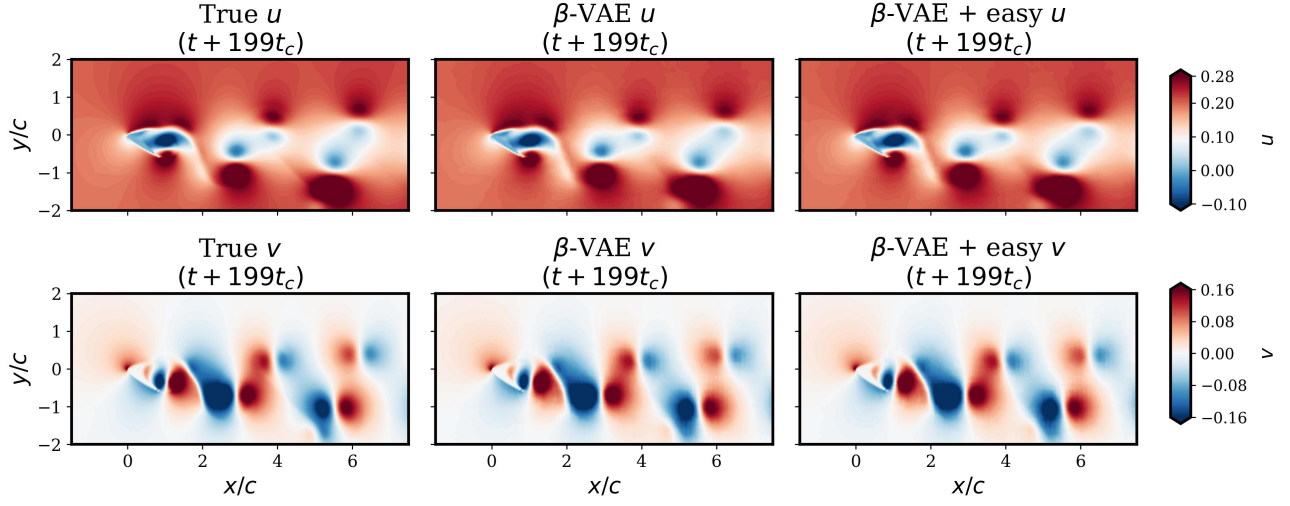
6 Supersonic jet at perfectly expanded mach number (M_j) = 1.19

The supersonic jet case at $M_j = 1.19$ provides a more stringent assessment of the β -VAE because the flow contains both coherent shear-layer motion and shock structures. The results show a jet with a sonic exit condition, with an exit pressure ratio of 1.26 at atmospheric pressure and temperature. In a supersonic jet, sharp gradients, wave interactions, shock-cell patterns, and shock-shear-layer interactions can appear in addition to the vortical structures observed in subsonic jets. These features make it more difficult to represent the flow using a low-dimensional model, because a small spatial error near a sharp gradient can lead to a large local reconstruction error.

The cumulative reconstructed energy for the supersonic jet is shown in Fig. 13. The β -VAE reconstruction energy increases with the number of latent variables and then gradually approaches a plateau. Compared with the subsonic jet case, the increase is less rapid, indicating that more latent variables are required to reach a similar reconstruction level for the present supersonic jet dataset. The POD reconstruction energy also increases with the number of modes, but it remains lower than the β -VAE over the same modal range. Therefore, based on the reconstructed-energy metric, the

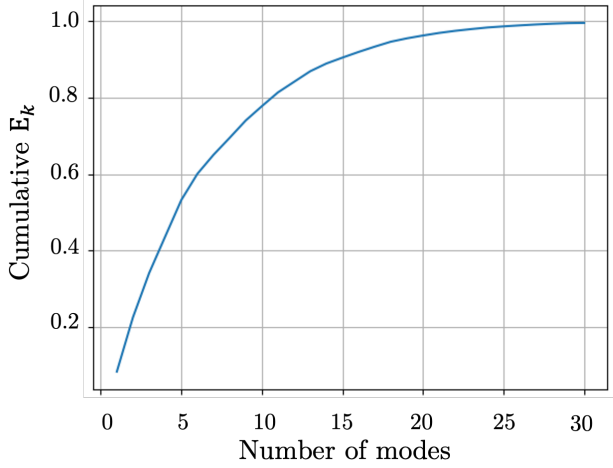


(a) $t + 65t_c$

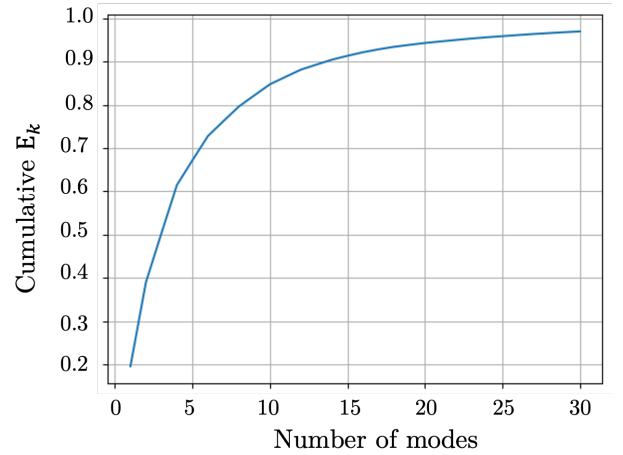


(b) $t + 199t_c$

Figure 9: True, reconstructed, and predicted fields for the airfoil case at $\alpha = 35^\circ$.

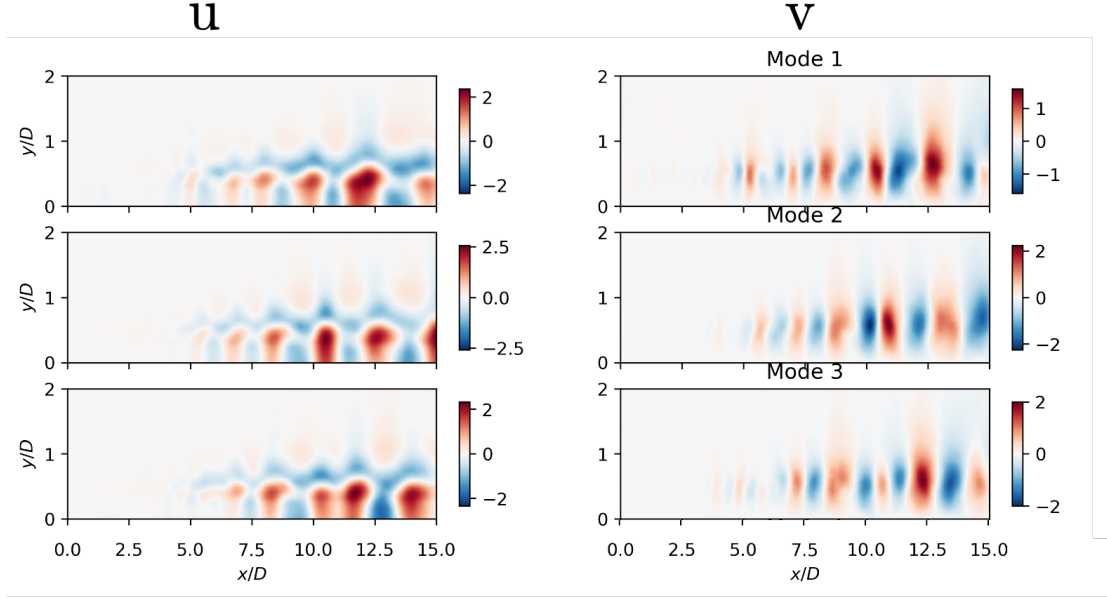


(a) β -VAE

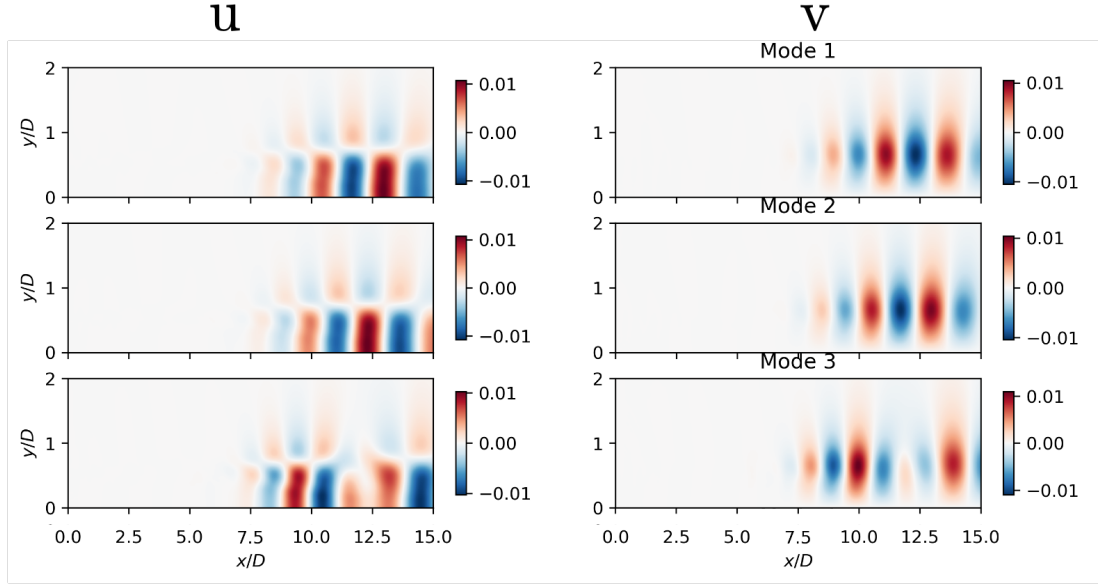


(b) POD

Figure 10: Cumulative reconstructed energy for the subsonic jet at mach number = 0.6.



(a) β -VAE modes



(b) POD modes

Figure 11: Spatial modes for the subsonic jet at $M = 0.6$

β -VAE provides a more compact representation than POD for the present supersonic jet case. However, the plateau in the β -VAE curve also shows that adding more latent variables does not lead to proportional improvement beyond a certain point.

The spatial modes shown in Fig. 14 further compare the β -VAE and POD representations for the supersonic jet. The β -VAE modes show localized positive and negative structures within the jet region, with noticeable variations in the downstream part of the domain. The POD modes also show spatial variations in the jet plume, but the structures are distributed differently across the retained modes. Based on the mode visualizations, both methods capture the organized spatial features of the supersonic jet, but the resulting modal representations differ. Together with the reconstructed-energy result in Fig. 13, the comparison shows that the β -VAE achieves a higher reconstructed energy than POD for the same number of retained modes in the present supersonic jet dataset. The prediction results in

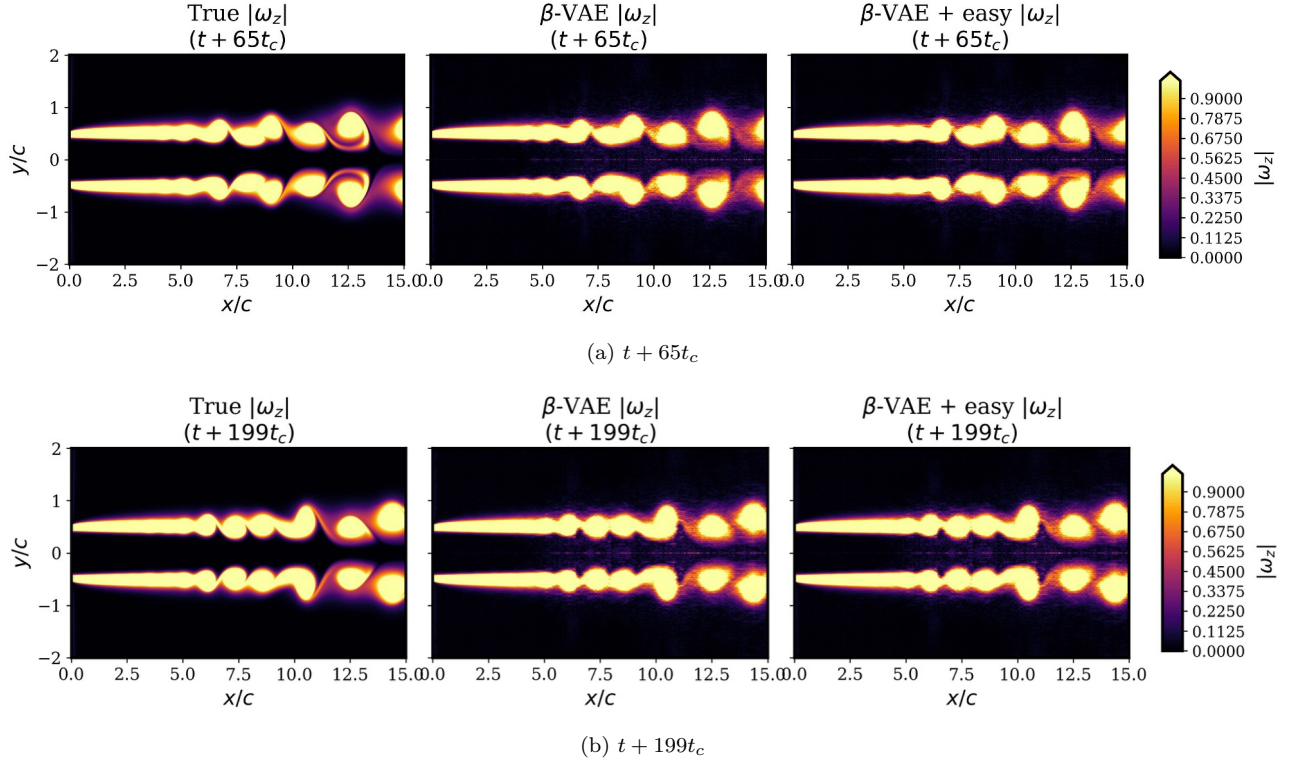


Figure 12: True, reconstructed from β -VAE, and transformer predicted fields for the subsonic jet at $M = 0.6$.

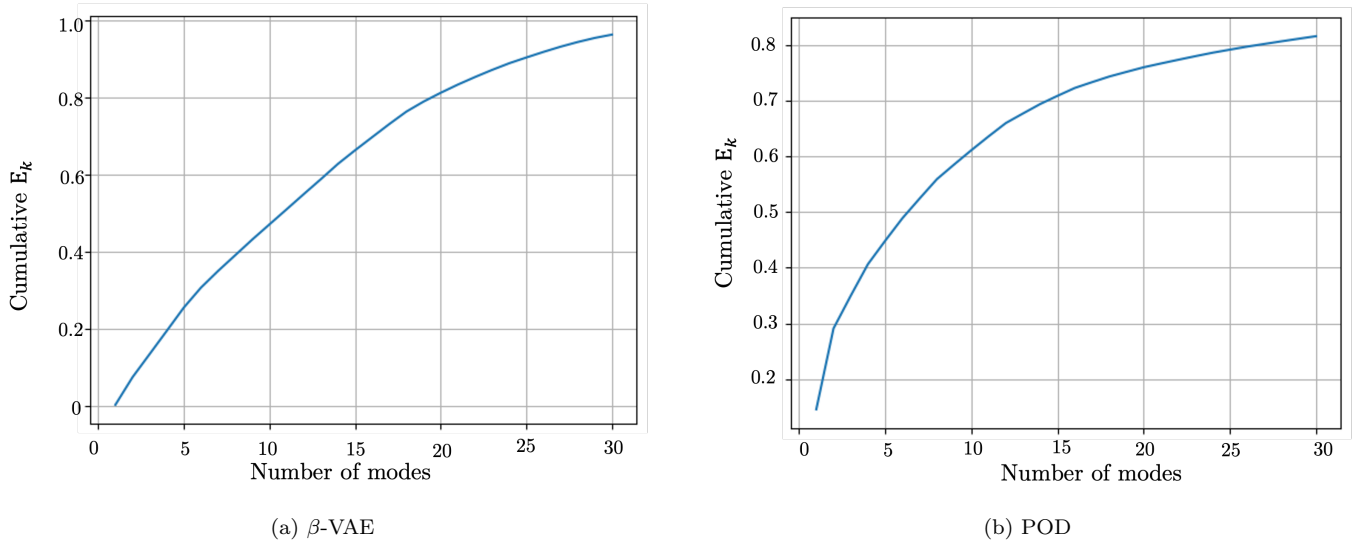
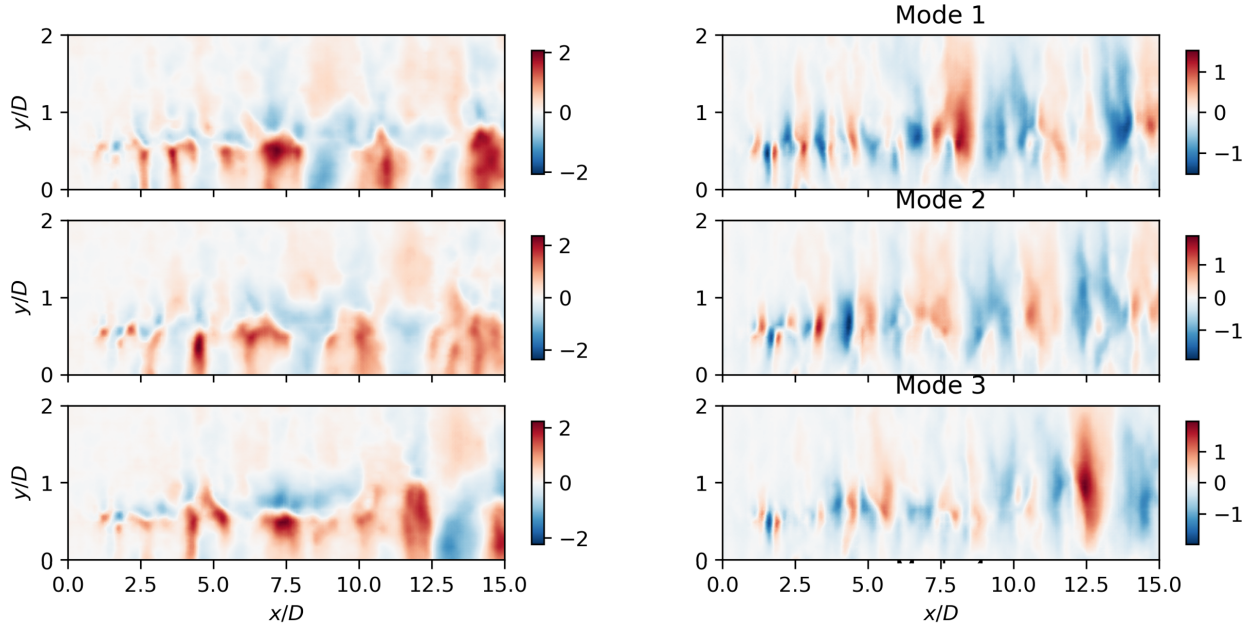
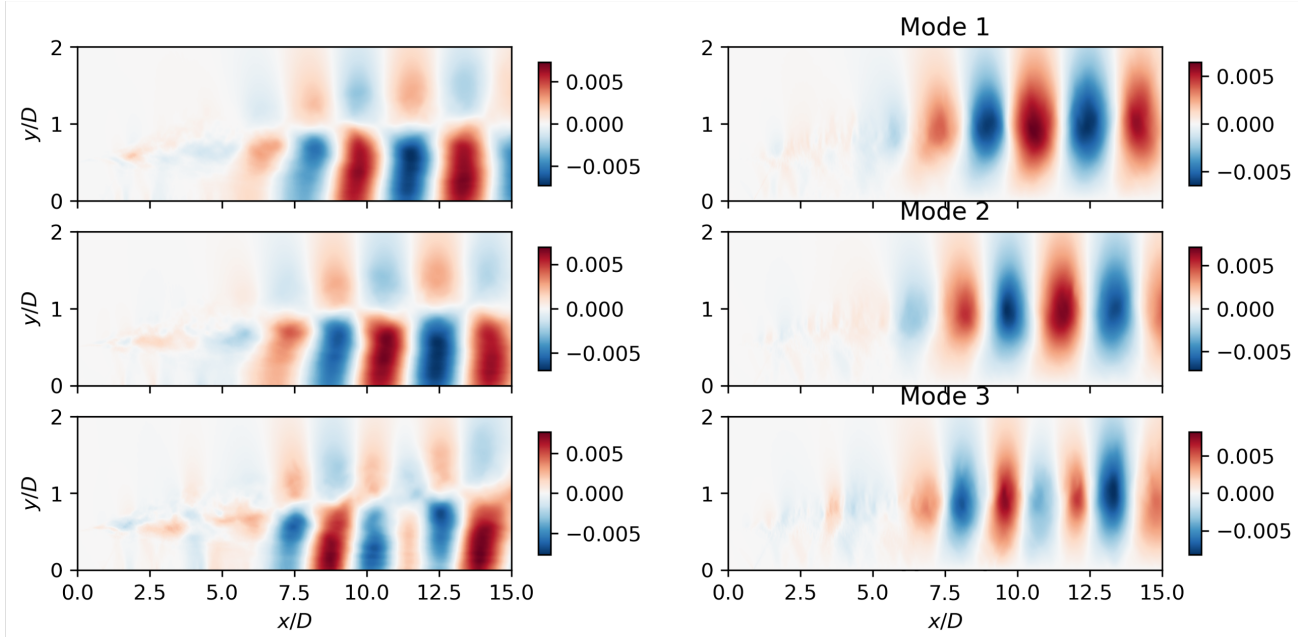


Figure 13: Cumulative reconstructed energy for the supersonic jet at perfectly expanded Mach number $M_j = 1.19$.



(a) β -VAE modes



(b) POD modes

Figure 14: Spatial modes for the supersonic jet at $M_j = 1.19$

Fig. 15 show the true vorticity magnitude field, the direct β -VAE reconstruction, and the β -VAE plus easy-attention prediction at $t + 65t_c$ and $t + 140t_c$. Both the reconstructed and predicted fields preserve the overall jet envelope and the approximate location of the main high-vorticity regions. This indicates that the latent representation captures the dominant large-scale features of the supersonic jet. However, some limitations are also observed. The true vorticity fields exhibit sharper, more filament-like structures, whereas the reconstructed and predicted fields appear smoother and more diffuse. Some local high-vorticity details are smeared, and the contrast between the jet core, shear layer, and surrounding region is reduced. Another important observation is that the β -VAE reconstruction and the β -VAE plus easy-attention prediction show similar types of error. This suggests that part of the error is associated with the decoder reconstruction capability, rather than only the temporal prediction model. Therefore, even if the latent-space predictor evolves the dominant dynamics, shock-related and small-scale structures cannot be recovered accurately

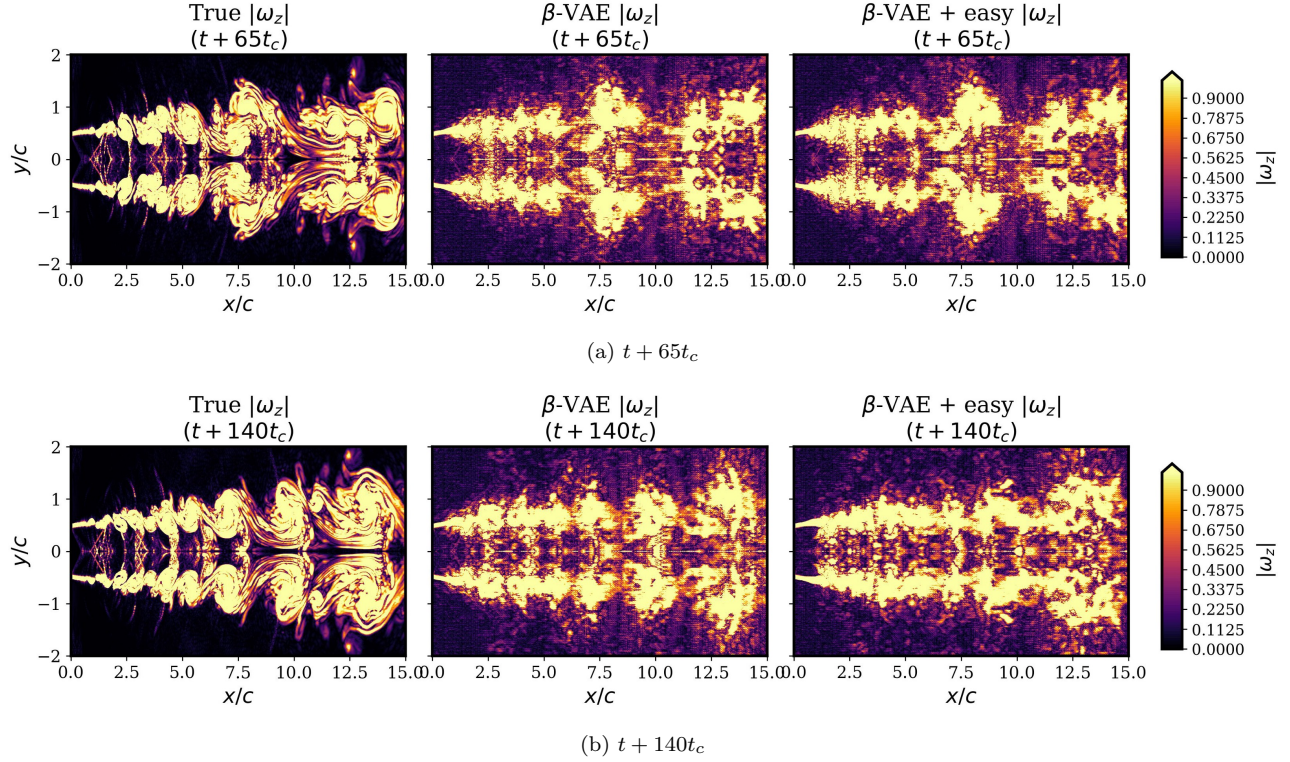


Figure 15: True, reconstructed from β -VAE, and transformer predicted fields for the supersonic jet at $M_j = 1.19$.

if they are not preserved by the decoder. The supersonic jet case, therefore, demonstrates both the usefulness and limitations of the current β -VAE framework. The model provides a compact representation compared with POD, but sharp gradients, shock-related structures, and localized vorticity are not fully preserved. For future shock-dominated flow reconstruction, the loss function should include additional structural information, such as Structural Similarity Index Measure (SSIM) loss [2, 3], gradient-based loss, or shock/edge-weighted reconstruction loss in addition to mean square error loss and Kullback-Leibler(KL) regularization.

7 Comparison and Conclusions

The present assessment compares the β -VAE and POD for flow cases of increasing complexity, including cylinder wake, airfoil wake, separated airfoil wake, subsonic jet, and supersonic jet. The objective is to evaluate whether the β -VAE can provide a compact nonlinear latent representation for future jet-impingement and plume-surface-interaction surface-loading prediction.

For the cylinder and low-angle airfoil cases, both POD and β -VAE perform well because the flow is dominated by organized wake dynamics. However, the β -VAE achieves comparable reconstruction using fewer latent variables, indicating a more compact representation. For the high-angle airfoil case, the advantage of the β -VAE becomes clearer because the separated wake is reconstructed more efficiently than with POD.

For the subsonic jet, the β -VAE provides a compact representation of the spatially developing jet and preserves the overall jet structure in the predicted fields. For the supersonic jet, the β -VAE still performs better than POD in terms of reconstructed energy, but the reconstructed and predicted fields are smoother than the true field. This shows that sharp gradients, shock-related structures, and localized vorticity are not fully preserved by the current decoder.

Overall, the assessment shows that the β -VAE is a suitable reduced-order framework for representing the dominant structures of the tested flow fields. However, for shock-dominated supersonic flows, reconstructed energy alone is not sufficient. Future work will include local error metrics and improved loss functions, such as Structural Similarity Index Measure (SSIM), gradient-based loss, or shock/edge-weighted loss, to better preserve sharp flow features before applying the framework to the full PSI surface-loading database.

References

- [1] Alberto Solera-Rico, Carlos Sanmiguel Vila, Miguel Gómez-López, Yuning Wang, Abdulrahman Almashjary, Scott TM Dawson, and Ricardo Vinuesa. β -variational autoencoders and transformers for reduced-order modelling

Table 1: Summary of reduced-order representation behavior.

Case	Flow behavior	β -VAE	POD
Cylinder	Periodic vortex shedding	Compact representation with few latent variables	Captures dominant shedding with first mode pair
Airfoil $\alpha = 10^\circ$	Organized wake	High reconstruction with few latent variables	Performs well but requires more modes
Airfoil $\alpha = 35^\circ$	Strong separated wake	Efficient nonlinear reconstruction	Energy spread over more modes
Subsonic jet, $M = 0.6$	Spatially developing jet	Higher reconstructed energy for the same number of variables	Lower reconstructed energy over the same modal range
Supersonic jet, $M_j = 1.19$	Compressible jet with shock structures	Compact but smoother reconstruction	Less compact representation

of fluid flows. *Nature Communications*, 15(1):1361, 2024.

- [2] Ziyang Liu, Renkun Han, Miao Zhang, Yang Zhang, Hao Zhou, Gang Wang, and Gang Chen. An enhanced hybrid deep neural network reduced-order model for transonic buffet flow prediction. *Aerospace Science and Technology*, 126:107636, 2022.
- [3] Chen Kong, Chenlin Zhang, Ziao Wang, Yunfei Li, and Juntao Chang. Efficient prediction of supersonic flowfield in an isolator based on pressure sequence. *AIAA Journal*, 60(5):2826–2835, 2022.